



DW_exp2

Base 2 Exponential (2^a)

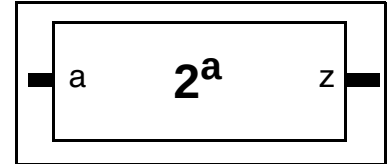
Version, STAR and Download Information: [IP Directory](#)

Features and Benefits

- Parameterized word width
- Supports up to 60 bits of precision

Revision History

Description



The DW_exp2 component takes an input *a* in fixed-point format and computes the function 2^a . The input must be in the range (0,1) and therefore the output is in the range (1,2) (normalized).

The number of bits used as input and output is defined by the *op_width* parameter. The input has *op_width* fractional bits. The output has one integer bit (always one) and *op_width* - 1 fractional bits.

The component implements the exponential function based on different algorithms for different ranges of precision. The selection of the algorithm is done automatically to deliver the best QoR.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
a	<i>op_width</i> bits	Input	Input data in the range (0,1)
z	<i>op_width</i> bits	Output	2^a in the range (1,2)

Table 1-2 Parameter Description

Parameter	Values	Description
op_width	2 to 60 ^a Default: 8	Word length of <i>a</i> and <i>z</i>
arch	0 or 2 Default: 2	Implementation selection ■ 0 = Area optimized ■ 1 = Speed optimized ■ 2 = Obsolete implementation retained for backward compatibility; will be removed in a subsequent release
err_range	1 or 2 Default: 1	Error range of the result compared to the infinitely precise result ■ 1 = 1 ulp ■ 2 = 2 ulps

a. The synthesis model fully supports this range, as does the Verilog simulation model in VCS, but the VHDL simulation model (in all simulators) and the Verilog simulation model in non-VCS simulators are limited to a range of 2 - 38.

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rtl	Implement using the Datapath Generator technology combined with static DesignWare components	DesignWare

Table 1-4 Simulation Model

Model	Function
DW02.DW_EXP2_CFG_SIM	Design unit name for VHDL simulation
dw/dw02/src/DW_exp2_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_exp2.v	Verilog simulation model source code

The truth table for the function implemented by this component when *op_width* = 4, *arch* = 2 and *err_range* = 1 is shown in Table 1-5. The values provided for eight bits of precision were truncated from a longer-precision value. The MS bit of the output vector is always 1. Also, observe that the error is always less than the weight of the LS bit position of the output; in the example, less than 2⁻³.

Table 1-5 Truth Table (*op_width* = 4, *arch* = 2, *err_range* = 1)

a(3:0)	a value (decimal)	z(3:0)	z value (decimal)	2 ^a (8 bits)
.0000	0.0	1.000	1.000	1.0000000
.0001	0.063	1.000	1.000	1.0000101
.0010	0.125	1.000	1.000	1.0001011
.0011	0.188	1.001	1.125	1.0010001
.0100	0.250	1.001	1.125	1.0011000
.0101	0.313	1.001	1.125	1.0011110
.0110	0.375	1.010	1.250	1.0100101
.0111	0.438	1.010	1.250	1.0101101
.1000	0.500	1.011	1.375	1.0110101
.1001	0.563	1.011	1.375	1.0111101
.1010	0.625	1.100	1.500	1.1000101
.1011	0.688	1.100	1.500	1.1001110
.1100	0.750	1.101	1.325	1.1010111
.1101	0.813	1.110	1.750	1.1100000
.1110	0.875	1.110	1.750	1.1101010

Table 1-5 Truth Table (*op_width* = 4, *arch* = 2, *err_range* = 1) (Continued)

a(3:0)	a value (decimal)	z(3:0)	z value (decimal)	2 ^a (8 bits)
.1111	0.938	1.111	1.875	1.1110101

Another parameter, *err_range*, can be used to relax the error boundary and get an implementation with slightly better QoR. The error does not exceed 2 ulps when *err_range* = 2, and it does not exceed 1 ulp when *err_range* = 1, except for the configurations with odd values of *op_width* >= 49, the maximum error is 2 ulps. The error of 1 ulp corresponds to $2^{-op_width+1}$.

The *arch* parameter controls implementation alternatives for this component. Different values result in different numerical behavior. You should experiment with the *arch* parameter to find out which value provides the best QoR for your design constraints and technology. Using *arch* = 0 (area optimized implementation) usually provides the best QoR for most time constraints.

To compute the exponential of a negative value in two's complement, and avoid the use of a reciprocal or division operation, the designer can use a combination of simple transformations. Assume a negative input *x*, with $|x| < 1$. We compute 2^x as:

$$2^x = \frac{2^x * 2}{2} = \frac{2^{x+1}}{2}$$

where *x* is in the proper range for DW_exp2. The value $a = (1 + x)$ is easily obtained from *x* by discarding the sign bit. The output of DW_exp2(*a*) is then shifted to the right to obtain the final output. For example: $x = -3/8 = 1.1010$, $a = 0.1010$, $2^a = 1.100$, and thus $2^x = 0.1100 = 0.75$. The value 2^x computed with 8 bits of precision would be 0.11000101.

Alternative Implementation of Base-2 Exponential with DW_lp_multifunc

The base-2 exponential operation can also be implemented by the DW_lp_multifunc component, which evaluates the value of base-2 exponential with 1 ulp error bound. There will be 1 ulp difference between the value from DW_lp_multifunc and the value from DW_exp2. Performance and area of the synthesis results are different between the DW_exp2 and the base-2 exponential implementation of the DW_lp_multifunc, depending on synthesis constraints, library cells and synthesis environments. By comparing performance and area between these alternatives, the DW_lp_multifunc provides more choices for better synthesis results. Below is an example of the Verilog description for the base-2 exponential of the DW_lp_multifunc. For more, see [DW_lp_multifunc](#).

```
DW_lp_multifunc #(op_width, 64) U1 (
    .a(a),
    .func(16'h0040),
    .z(z),
    .status(status)
);
```

Related Topics

- [Application Specific – Data Integrity Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_Foundation_comp_arith.all;

entity DW_exp2_inst is
    generic (
        inst_op_width : INTEGER := 8;
        inst_arch : INTEGER := 2;
        inst_err_range : INTEGER := 1
    );
    port (
        inst_a : in std_logic_vector(inst_op_width-1 downto 0);
        z_inst : out std_logic_vector(inst_op_width-1 downto 0)
    );
end DW_exp2_inst;

architecture inst of DW_exp2_inst is

begin

    -- Instance of DW_exp2
    U1 : DW_exp2
        generic map (
            op_width => inst_op_width,
            arch => inst_arch,
            err_range => inst_err_range
        )
        port map (
            a => inst_a,
            z => z_inst
        );

end inst;

-- pragma translate_off
configuration DW_exp2_inst_cfg_inst of DW_exp2_inst is
for inst
end for; -- inst
end DW_exp2_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_exp2_inst( inst_a, z_inst );

parameter inst_op_width = 8;
parameter inst_arch = 2;
parameter inst_err_range = 1;

input [inst_op_width-1 : 0] inst_a;
output [inst_op_width-1 : 0] z_inst;

// Instance of DW_exp2
DW_exp2 #(inst_op_width, inst_arch, inst_err_range) U1 (
    .a(inst_a),
    .z(z_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	Updated date and version; no technical changes
December 2023	DWBB_202312.0	Updated date and version; no technical changes
June 2023	DWBB_202212.4	Removed note about ulp error range in Table 1-2 on page 1 and clarified ulp details just below Table 1-5 on page 2
January 2020	DWBB_201912.1	Corrected port names for DW_lp_multifunc in “Alternative Implementation of Base-2 Exponential with DW_lp_multifunc” on page 3
June 14, 2019	DWBB_201903.2	Removed reference to minPower library in “Alternative Implementation of Base-2 Exponential with DW_lp_multifunc” on page 3
March 2019	DWBB_201903.0	- Clarified value ‘2’ of the arch parameter in Table 1-2 on page 1; this value will be obsoleted in a subsequent release - For STAR 9001459848, removed the ‘str’ implementation from Table 1-3 on page 2 because the QoR benefit is now provided in the ‘rtl’ implementation
July 2018	DWBB_201806.1	- For STAR 9001366624, in Table 1-2 on page 1, clarified the range of <i>op_width</i> for the VHDL simulation model (in all simulators) and the Verilog simulation model for non-VCS simulators. - Added this Revision History table and the document links on this page

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